

	Case Name: Fontein	Sector	Construction (Civil)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
Submitted by	Pieter Van Schoors		One of nine std. scenarios
Date	2013		User defined distributions
File Name	C2018-10 Fontein	Project Control	Automatic tracking
			Tracking based on user input

1. Project description

Project authenticity

The construction and installation of a fountain system, including structural works, hydraulic components, and finishing elements.

The project consists of activity, resource, and cost data obtained directly from the actual project execution.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	43	Serial/Parallel (SP)	58%
Planned Duration (PD)	100d	Activity Distribution (AD)	62%
Budget At Completion (BAC)	115,115.24€	Length of Arcs (LA)	4%
Renewable Resources	-	Topological Float (TF)	39%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	6	15	3.10
CRI-rho	45	13	-2.01
CRI-tau	82	35	-1.57

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	43	50	0.30
SI	48	46	0.20
SSI	4	15	6.27
CRI-r	12	15	5.18
CRI-rho	13	16	4.32
CRI-tau	14	20	3.33

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	0	0	N/A
CRI-tau	0	0	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	N/A	N/A
PV - SPI	N/A	N/A
PV - SCI	N/A	N/A
ED - 1	N/A	N/A
ED - SPI	N/A	N/A
ED - SCI	N/A	N/A
ES - 1	N/A	N/A
ES - SPI(t)	N/A	N/A
ES - SCI(t)	N/A	N/A

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	N/A	N/A
CPI	N/A	N/A
SPI	N/A	N/A
SPI(t)	N/A	N/A
SCI	N/A	N/A
SCI(t)	N/A	N/A
0.8 CPI + 0.2 SPI	N/A	N/A
0.8 CPI + 0.2 SPI(t)	N/A	N/A

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over **24** tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0	-5406.17	-7.95	1	0.95	0.92	1
std dev	0	8788.47	9.48	1	0.08	0.10	0
final	0	0	-20.88	1	1.00	0.83	1

2.3.3.2. Time forecasting

PD	100d	Real Duration	120d	Late/early	20%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	208.46	7.42	13.09	12.95
PV - SPI	209.88	9.92	13.51	11.77
PV - SCI	209.88	9.92	13.51	11.77
ED - 1	210.50	8.05	11.25	11.25
ED - SPI	212.13	10.42	11.63	9.90
ED - SCI	212.13	10.42	11.63	9.90
ES - 1	211.92	9.43	10.07	10.07
ES - SPI(t)	213.88	11.73	10.87	8.44
ES - SCI(t)	213.88	11.73	10.87	8.44

2.3.3.3. Cost forecasting

BAC	115,115.24€	Real Cost	115,115.24€	Over/under budget	0%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	115115.23	0.00	0.00	0.00
CPI	115115.23	0.00	0.00	0.00
SPI	116736.98	3086.17	0.00	0.00
SPI(t)	116936.96	3038.41	0.00	0.00
SCI	116736.98	3086.17	0.00	0.00
SCI(t)	116936.96	3038.41	0.00	0.00
0.8 CPI + 0.2 SPI	115387.50	505.44	0.00	0.00
0.8 CPI + 0.2 SPI(t)	115420.14	500.80	0.00	0.00